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Abstract

Target leakage, adjacent-window overlap, participant mixing and evaluation on familiar driving routes can all inflate reported performance in driver-drowsiness detection. This study is a leakage-resistant secondary analysis of two author-supplied multimodal dataset families. The Driver Monitoring and Physiological Sensing (DMPS) workbooks include 225,000 observations from 15 drivers on highway, rough-terrain and dense-urban routes; the skin-conductance (SC) workbook includes 87,430 overlapping 30-s windows from 20 participants. The variables that were directly measured from the events and alternate encodings of the drowsiness target were removed prior to modeling. Logistic regression, pooled histogram-based gradient boosting (HGB) and calibrated quality-weighted modality fusion (CQMF) were assessed in the context of random-row, within-sequence time-blocked, driver-held-out and route-held-out protocols. Probabilities were calibrated on separate calibration subsets and evaluated with AUROC, AUPRC, F1, Matthews correlation coefficient, Brier score, expected calibration error, and missing-sensor stress tests. Under five-fold driver-held-out validation, pooled HGB achieved AUROC 0.9990 ± 0.0003, AUPRC 0.9819 ± 0.0041, F1 0.9342 ± 0.0094, MCC 0.9319 ± 0.0098, Brier score 0.00355 ± 0.00039, and expected calibration error 0.00149 ± 0.00121. If the whole route was left out, mean AUPRC dropped to 0.9686 and mean sensitivity fell to 0.7933, revealing a route-transfer penalty that was not seen when using random splitting. CQMF did not outperform pooled HGB (driver-held-out AUPRC 0.9062), indicating that separating the modalities lost useful cross-modal interactions. When

sensor values were missing at random, pooled HGB remained the stronger model, although its AUPRC fell from 0.9819 to 0.6630 at 50% missingness. Held-out separability remained high in the SC sensitivity analysis after participants were separated; however, the supplied binary label was deterministically linked to KSS and raw acquisition streams were unavailable, which limits its value as external evidence. The results help to validate calibrated pooled multimodal modeling and illustrate the need for leakage auditing, route-held-out validation, and transparent provenance as prerequisites for credible safety claims.

Keywords: driver drowsiness detection; multimodal sensing; data leakage; cross-driver validation; route generalization; probability calibration; gradient boosting

1. Introduction

Drowsy driving impairs vigilance, reaction times, decision-making and vehicle control. Detection systems are thus physiological, behavioural, visual and vehicle-dynamic. The present research team has previously developed from the wearable biological monitoring/DAI paradigm [1] to the vision-based fatigue recognition paradigm [2] to the adaptive skin-conductance modeling paradigm [3] to the context-aware driver–vehicle–environment paradigm [4]. This evolution is a natural foundation for multi-modal detection, but brings up an unanswered methodological question: whether reported performance can be seen as valid when tested on drivers and routes that were not involved in the development of the model.

Changes in physiology can precede behavioural changes, however there are confounders for each measurement. Posture, stress, boredom and relaxation, as well as sleepiness, affect heart-rate variability [5,21]. Skin conductance is another index of sympathetic arousal that can be used in wearable detection [6] but varies from person to person and is affected by motion and environment. Camera-based methods that track facial and ocular behaviour have the advantage of being non-contact and intuitive, but can suffer from domain shift due to illumination, occlusion, camera pose and driver identity [2,9]. To take advantage of complementary evidence, multimodal systems have been developed [7,8,10,16]; and over the past years, public datasets have become more and more focused on synchronization and continuous sessions [11].

As well as the classifier, the design of the evaluation is important. When splitting rows from a time series containing overlapping rows, random sampling may end up placing near duplicate rows in both the training and test sets. The potential for subject-specific baselines in participant mixing can enable models to perform well at the task without learning anything useful except a signal specific to the task at hand, while target proxies – variables generated using the same annotation rule as the outcome – can help models look perfect without learning anything of value. The recently published reproducibility study of vehicle-dynamics-based drowsiness detection found significant material differences between published and reproduced performance and highlighted leakage and non-standard evaluation as key issues [12]. Class imbalance is the main factor that accentuates these problems; in the current route information, the prevalence of drowsy conditions varies between 1.68% and 5.06% of the overall population, thus the increase in accuracy and even AUROC could result in large variations in precision or sensitivity of the positive event. Therefore, it is important to use precision-recall analysis [19].

Another deployment gap is probability calibration. Observations must not only rank the safety systems, but they must also deliver confidence estimates that are understandable for the selection of a threshold, escalation and human–machine interaction. Even when discrimination is high, modern classifiers can remain poorly calibrated [18]. Calibration should be assessed on data that is not used for model fitting nor final testing and should be measured by appropriate scoring rules like the Brier score [20] and reliability measures.

The need for a leakage-resistant secondary analysis of the DMPS route workbooks and a SC workbook is addressed in this study. The contributions are: (i) an explicit audit of direct event leakage, alternate target encodings, overlap, prevalence, and temporal provenance; (ii) a controlled comparison of random-row, time-blocked, driver-held-out, and route-held-out validation; (iii) independent probability calibration and threshold selection; (iv) comparison of linear, pooled nonlinear, and quality-weighted modality-fusion models; (v) modality ablation and missing-sensor stress testing; and (vi) a cautious interpretation that separates internal discrimination from claims of real-world readiness. This study addresses same-time (contemporaneous) drowsiness detection rather than early warning, because the supplied route workbooks do not provide a validated pre-event prediction horizon.

2. Related work

Physiological and vehicle-based driver monitoring is less dependent on ambient lighting and an unobstructed view of the face. The previous intelligent vehicle platform connected HR, BP, breathing rate, SpO_2, grip, foot motion and vehicle parameters, with 15 subjects in 3 routes [1]. Later research confirmed that the combination of physiological signals can outperform single signals, but with performance varying depending on the ground-truth definition and validation scheme [5,7]. Minimally-intrusive sensing has been demonstrated with these two models of wrist-based skin conductance [6] and blood-volume-pulse [13,21] sensors.

Deep-ATM DL-LSTM unified tonic and phasic SC with adaptive thresholds gave an accuracy of 96.4% and an AUC of 0.978 for four drowsiness levels [3]. However, its 30-s windows overlapped by 50%, and the reported 70/15/15 stratified split did not establish that participants were disjoint before segmentation. In the present study, therefore, the provided SC workbook is considered to be a sensitivity dataset and the analysis is repeated without including KSS, drowsiness level, drowsiness state, and continuous drowsiness score as predictors.

VigilEye employed a total of 68 facial landmarks and traditional classifiers to detect fatigue with 94% accuracy and observed that its performance decreased when the camera was close to the driver or the lighting was low [2]. Multi-modal vision–physiology models have been recently reported with significant improvement with feature coupling or all-weather fusion [9,10]. Multi-sensor driver monitoring can also use facial, physiological and vehicle signals for prediction [8]. Public datasets like the 60 subject early drowsiness video dataset [15] and the 19 subject UL-DD dataset [11] augment the reproducibility and contain continuative or synchronized streams. However, metrics reported for each dataset are not directly comparable as there are differences in labels, sampling units, participants, class balance, and split rules.

Most papers that develop models optimize the architecture in one data set. The present work instead seeks to determine which claims are still supportable once leakage variables are removed and the evaluation unit is shifted from rows to drivers or routes. This emphasis is aligned with recent worries on reproducibility [12] and the requirement to report AUPRC and calibration for rare safety occasions [18–20]. Table 1 compares the study to the four source papers and to some recent literature.

Table 1. Research progression and position of the present study.

Study	Modalities	Evaluation data	Reported result	Issue addressed here
Ponnan et al. [1]	Physiology, wearables, vehicle	15 drivers; 3 routes	72.54% profile matching	No driver/route-held-out predictive benchmark

Theivadas and Ponnan [2]	Facial landmarks	Camera scenarios	94% efficiency	Lighting and distance sensitivity; split unit not reported
Theivadas and Ponnan [3]	SC and context	20 drivers; overlapping 30-s windows	Accuracy 96.4%; AUC 0.978	Participant-disjoint segmentation not established
Theivadas and Ponnan [4]	Physiology, driver, vehicle, context	30 drivers; 300 sessions	Safety accuracy 88.3%; AUC 0.923	Different behavioural target; not a leakage audit
Hasan et al. [7]	Multimodal physiology	Multiple validation methods	Explainable ML validation	Dataset-specific; no route-held-out analysis
Cao et al. [10]	EEG, ECG, facial images	DROZY multimodal data	F1 98.38%	Different modalities and fatigue protocol
Bodaghi et al. [11]	Video, biometrics, grip, telemetry	19 participants; continuous sessions	Public multimodal descriptor	Dataset paper; limited benchmark comparability
Nakagama et al. [12]	Vehicle dynamics	Public-data reproduction	Best accuracy 88%	Identified leakage/reproducibility concerns
Present study	Physiology, wearable behaviour, vehicle context	225,000 route rows + 87,430 SC windows	Driver-held-out HGB AUPRC 0.9819	Secondary workbooks; raw streams unavailable

3. Materials and methods

3.1 Study design and data sources

This was a secondary analysis undertaken on four author supplied Excel workbooks in a retrospective manner. Three DMPS workbooks were related to highway, rough-terrain, and dense-urban routes. There were 15 sheets (Driver_01 to Driver_15) for each workbook containing 5,000 observations and 28 original columns. Concatenation produced 225,000 rows. Identifiers for drivers were consistent between the routes, and were therefore used as the clustering unit. The dichotomous response variable was Event_Drowsiness. Event_Sudden_Brake was kept as only a descriptive auditor and is not being used as a predictor or co-target.

The SC workbook included a summary sheet and 20 sheets for the participants (P01–P20), which resulted in 87,430 30-s windows with 50% nominal overlap. The participants were 14 men and 6 women ranging in age from 24 to 50 years and totaling 811.65 driving hours were recorded. Highway-day, highway-night, urban-day, urban-night, rain, and frost were all included in the workbook. The main SC sensitivity measure was Binary_Drowsiness (alert/drowsy). The KSS is a validated subjective sleepiness rating scale [14] but, due to

the binary outcome being deterministically linked to KSS in the provided workbook, KSS was not included in the predictors. Table 2 summarises the resulting sample sizes and label distributions for all four workbooks, and Figure 1 shows the complete analysis workflow, from dataset audit and leakage exclusion through model fitting, calibration and protocol-specific evaluation.

Table 2. Dataset characteristics and label distributions.

Dataset	Drivers/ participants	Samples	Drowsiness positive	Sudden-brake positive	Structure
DMPS highway	15	75,000	3,795 (5.06%)	1,528 (2.04%)	No missing cells
DMPS rough terrain	15	75,000	2,627 (3.50%)	4,165 (5.55%)	No missing cells
DMPS dense urban	15	75,000	1,261 (1.68%)	5,869 (7.83%)	No missing cells
DMPS total	15	225,000	7,683 (3.41%)	11,562 (5.14%)	Same drivers across routes
SC workbook	20	87,430	63,152 (72.23%)	Not applicable	30-s windows; 50% overlap

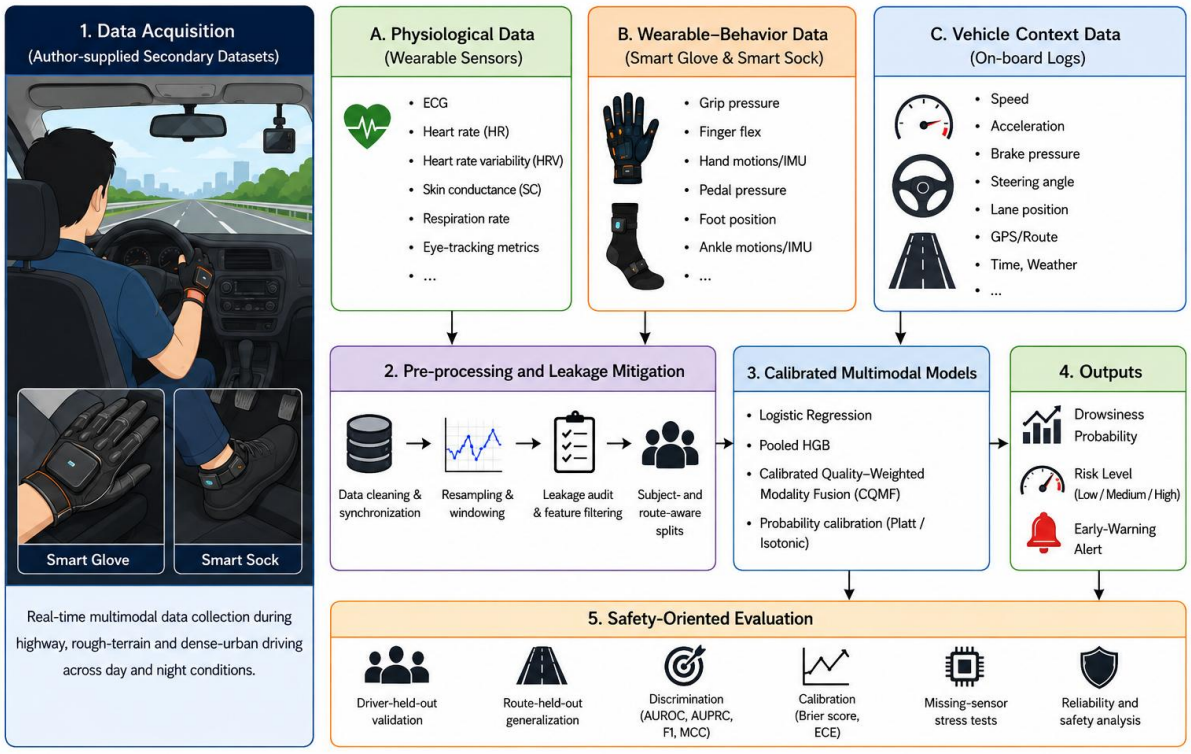


Fig. 1. Leakage-resistant multimodal driver-drowsiness analysis framework.

Figure 1. Leakage-resistant analysis workflow. The workflow separates dataset audit, model fitting, probability calibration, and final safety-oriented evaluation.

3.1.1 Dataset provenance, ethical approval and informed consent

The datasets analysed in this study were derived from the authors' previously published experimental studies and were supplied to the present research team as de-identified Excel workbooks. The DMPS workbooks originated from the driver-monitoring study reported by Ponnan et al. [1], which involved 15 willing participants assessed using smart-glove and smart-sock sensors across three driving-route conditions. Before participation, the volunteers signed a written agreement confirming their willingness and ability to participate in the experiment. The skin-conductance workbook originated from the Deep-ATM DL-LSTM study reported by Theivadas and Ponnan [3], which involved 20 professional drivers and was approved by the Ethics Review Committee of Vel Tech Rangarajan Dr Sagunthala R&D Institute of Science and Technology under approval number VT/ERC/2022-104.

All procedures associated with the original skin-conductance study were conducted in accordance with the relevant institutional guidelines and regulations and the Declaration of Helsinki. Written informed consent was obtained from all participants before participation, including consent for publication of anonymized data. The present study involved only retrospective secondary analysis of de-identified tabular workbooks and involved no new participant recruitment, intervention or collection of identifiable information.

3.2 Leakage and provenance audit

All variables were checked for face and item-to-total correlation prior to modeling. Both Brake_Intensity and Brake_Pressure_Socks separated Event_Sudden_Brake perfectly (univariate AUROC 1.000). These variables were not antecedents and were not used for drowsiness modeling, along with Event_Sudden_Brake. Additional columns, such as Sample_ID and timestamp, were also removed to avoid the model taking advantage of the row identity or the journey process. The drowsiness outcome was not forecasted in the future.

Drowsiness_State, Drowsiness_Level, KSS_Score, Binary_Drowsiness, and Drowsiness_Score_0_100 encoded the latent annotation of drowsiness in the SC workbook. Applying any of the alternate encodings to the input as such would therefore be a type of 'leakage' of the target; the AUROC for the binary label was 1.000 in the case of KSS. All other alternate targets were not included. Between 15-s-step consecutive windows, the binary label intermittently switched from 39.69% to another value in the participant sheets. These changes are not representative of a natural progression of drowsiness, but are fast and unpredictable. The SC analysis is, therefore, presented as a workbook sensitivity analysis rather than as an indication of the temporal prediction that occurs at an early stage.

Table 3. Predictor groups and leakage exclusions.

Group	Variables	Decision
Physiological (6)	Heart rate, systolic and diastolic BP, breathing rate, SpO_2, skin conductance	Included
Wearable behaviour (9)	Grip pressure, hand contact, hand movement, clutch and accelerator pressure, leg movement x/y, foot position, accelerometer ADC	Included
Vehicle context (6)	Wheel alignment, battery charge, engine RPM, cabin temperature, fuel consumption, indicator usage	Included
Direct event/identity (6)	Sample ID, time, distance, sudden-brake label, brake intensity, brake-pressure socks	Excluded

SC physiological/context (33 encoded)	18 SC descriptors plus one-hot driving condition, environment, lighting, and weather	Included in SC sensitivity analysis
SC target proxies (4)	Drowsiness state, drowsiness level, KSS, continuous drowsiness score	Excluded

3.3 Preprocessing

There were no missing values in the DMPS workbooks. After the leakage audit 21 predictors remained. The parameters estimated from the fitting subset to standardise logistic regression inputs are different from those that would be obtained if the full set of variables were used. HGB did not need scaling and was able to deal with nonlinear relationships and missing values as they are. Sample weights were computed within each of the fitting subsets to be class balanced. There was no random oversampling and/or synthetic minority generation.

Standardisation of the logistic-regression inputs uses only the location and scale estimated on the fitting subset, whose n_{fit} rows are indexed by i, so that for predictor j

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \tag{1}$$

where μ_j and σ_j are the mean and standard deviation of predictor j computed over the n_{fit} fitting rows only; the same constants are then applied unchanged to the calibration and test rows.

Eighteen continuous SC descriptors were chosen for the SC analysis: mean, standard deviation, number of responses, response amplitude, response frequency, area of the SC curve, rise time, decay time, and recovery time; spectral power, sample entropy, tonic level, phasic amplitude, derivative moments, skewness, kurtosis and number of non-specific responses. The driving condition, environment, lighting and weather were one-hot encoded, resulting in 33 input columns. All target proxies were excluded from the analysis, as were segment duration, overlap, session duration, and participant ID and timestamp.

3.4 Models

There were three models to compare. First, logistic regression, using class weights, was used to obtain a linear baseline. Secondly, pooled HGB learned all 21 DMPS predictors together. Gradient Boosting is an ensemble of fitting a sequence of trees to minimize a differentiable loss [17]. The implementation was provided with 19 leaf nodes per tree, minimum leaf size 60, L2 regularization 1.0, early stopping, learning rate 0.07 and at most 110 boosting iterations. The settings were determined prior to the final evaluation.

For a standardised predictor vector $x \in \mathbb{R}^d$ with d = 21, the logistic baseline models the drowsiness posterior as

$$\hat{p}(x) = \sigma\left(\beta_0 + \beta^{\mathrm{T}} x\right), \quad \sigma(z) = \frac{1}{1+e^{-z}} \tag{2}$$

and the coefficients are obtained by minimising the class-weighted, L2-penalised log loss over the fitting subset,

$$L(\beta_0, \beta) = -\sum_i w_i [y_i \ln \hat{p}_i + (1 - y_i) \ln(1 - \hat{p}_i)] + \frac{1}{2C} \beta^{\mathrm{T}} \beta \tag{3}$$

where the sample weight w_i depends only on the class label y_i and follows the balanced rule

$$w_c = \frac{N}{2N_c}, \quad c \in \{0, 1\} \tag{4}$$

with N_c the number of fitting rows of class c. No oversampling or synthetic minority generation was used, so Eq. (4) is the only mechanism compensating for the 3.41% positive prevalence.

The pooled HGB model instead builds an additive ensemble of M regression trees in the logit space,

$$F(x) = F_0 + \nu \sum_{m=1}^{M} h_m(x), \quad \hat{p}(x) = \sigma\big(F(x)\big) \tag{5}$$

where ν is the learning rate and each tree h_m is fitted to the second-order expansion of the log loss ℓ around the current ensemble F_{m-1},

$$h_m = \arg\min_h \sum_i w_i \left[g_i h(x_i) + \frac{1}{2} v_i \big(h(x_i)\big)^2 \right] + \lambda \Omega(h) \tag{6}$$

where g_i and v_i are the first and second derivatives of ℓ with respect to F evaluated at F_{m-1}, and $\Omega(h)$ is the leaf-weight penalty with strength λ.

The hyperparameters were fixed before the final evaluation at $M \leq 110$, $\nu = 0.07$, at most 19 leaves per tree, a minimum of 60 samples per leaf, and $\lambda = 1.0$, with early stopping on an internal split of I_{fit}. Histogram binning of the continuous predictors and native routing of missing values remove the need for the standardisation of Eq. (1).

Third, CQMF evaluated the robustness improvement by explicit modality separation. Physiological, wearable-behaviour and vehicle-context features were fitted to independent HGB learners. The log loss was minimized for non-negative fusion weights that sum to one to find the optimal weights on the first half of the calibration subset. The weights were then used to average the retained probabilities. For simulated missingness, each weight was multiplied by the observed fraction of its modality for the sample and then re-normalized. This design provided a direct test to the widely held notion that quality gating for each modality would inevitably be better for multimodal drowsiness detection.

Let $x = (x_{phys}, x_{wear}, x_{veh})$ partition the predictor vector into the three modality blocks of Table 3, and let $\hat{p}_k(x_k)$ be the probability produced by the HGB learner of block k. The fused probability is the convex combination

$$\hat{p}_{\text{fus}}(x) = \sum_k \alpha_k \hat{p}_k(x_k), \quad k \in \{\text{phys}, \text{wear}, \text{veh}\} \tag{7}$$

with the weight vector α restricted to the unit simplex $\Delta = \{ \alpha : \alpha_k \geq 0, \Sigma_k \alpha_k = 1 \}$. The weights are obtained by minimising the log loss on the first half of the calibration subset,

$$\alpha^* = \arg\min_{\alpha \in \Delta} - \sum_i \big[y_i \ln \hat{p}_{\text{fus}}(x_i) + (1 - y_i) \ln\big(1 - \hat{p}_{\text{fus}}(x_i)\big) \big] \tag{8}$$

Under simulated missingness, let $q_k(i) \in [0,1]$ be the observed fraction of the features of modality k for test sample i. The quality-gated weights are rescaled per sample and re-normalised as

$$\tilde{\alpha}_k(i) = \frac{\alpha_k^* q_k(i)}{\sum_j \alpha_j^* q_j(i)} \tag{9}$$

so that $\Sigma_k \tilde{\alpha}_k(i) = 1$ for every sample, and a fully unobserved modality receives zero weight without any refitting.

The Platt scaling was applied for each model and each fold, using the data that was not used for the fitting of the classifiers [18]. The second half of the calibration subset was used by CQMF to perform Platt scaling, following estimation of fusion weights using the first half of the subset. The operating Threshold (OT) for the binary classification was chosen in the calibration set to maximize the Matthews correlation coefficient (MCC). No test labels were used to inform scaling, weights or thresholds, with the test subset used only for final evaluation.

Let f(x) denote the uncalibrated decision score of any of the three models. Platt scaling fits two scalars A and B on the held-aside calibration subset,

$$\hat{p}(x) = \frac{1}{1+\exp(A\,f(x)+B)} \tag{10}$$

and the operating threshold is selected on the same calibration subset by maximising the Matthews correlation coefficient of Eq. (15),

$$\tau^{*} = \underset{\tau \in (0,1)}{\arg\max}\, \mathrm{MCC}(\tau) \tag{11}$$

so that $\hat{y}_i = 1$ if $\hat{p}_i \geq \tau^*$ and 0 otherwise. Because A, B, α^* and τ^* are all estimated without any access to test rows, the test subset contributes only to the final reported metrics. Algorithm 1 states the complete leakage-resistant procedure.

Algorithm 1. Leakage-resistant fitting, fusion, calibration and evaluation.

Input: predictor matrix X, label vector y, driver identifiers g, route identifiers r, protocol $\Pi \in$ {random, time-blocked, driver-held-out, route-held-out, participant-held-out}

Output: calibrated test probabilities $\hat{p}_{test}$ and metric set M

Drop all direct event, identity and target-proxy columns from X (Table 3), leaving d = 21 predictors

Partition indices into (I_{fit}, I_{cal}, I_{test}) according to Π, with $I_{fit} \cap I_{cal} \cap I_{test} = \emptyset$ and no driver, route or overlapping window shared across the partitions

Estimate μ_j, σ_j on I_{fit} only and standardise by Eq. (1) ▷ logistic regression only

Compute class weights w_i from the label frequencies of I_{fit} by Eq. (4)

for each modality k ∈ {physiological, wearable, vehicle} do

    Fit HGB learner $\hat{p}$k on I_{fit} using only the columns of modality k, Eqs. (5)–(6)

end for

Split I_{cal} into two disjoint halves $I_{cal}^{(1)}$ and $I_{cal}^{(2)}$

Solve Eq. (8) on $I_{cal}^{(1)}$ for the simplex-constrained fusion weights α^*

Fit pooled HGB and weighted logistic regression on I_{fit}, Eqs. (2)–(3), (5)–(6)

for each model m ∈ {logistic, HGB, CQMF} do

    Fit Platt parameters (Am, Bm) by Eq. (10) on I_{cal} for logistic and HGB, and on $I_{cal}^{(2)}$ for CQMF

    Select τ^*m on the same calibration data by Eq. (11)

end for

if missing-sensor stress test then mask a fraction $\rho \in \{0.10, 0.20, 0.30, 0.50\}$ of test cells at random

    Recompute per-sample quality weights $\tilde{\alpha}$k(i) by Eq. (9); do not refit any model

end if

Predict $\hat{p}_{test}$ on I_{test}, threshold at τ^*m, and evaluate Eqs. (12)–(18)

Aggregate fold results as mean ± SD and obtain driver-clustered bootstrap intervals (200 replicates, 15 clusters)

return $\hat{p}_{test}$, M

3.5 Validation protocols

Table 4 defines the four DMPS partitioning rules and the SC participant-held-out rule, together with the specific form of optimism each rule is designed to expose. The four DMPS protocols differ only in how the fitting, calibration and test indices are formed; the predictor set, the hyperparameters and the calibration procedure are identical throughout, so any difference in reported performance is attributable to the partitioning rule alone rather than to model capacity.

Figure 2 contrasts the four schemes visually. In every protocol the calibration block is disjoint from both the fitting and the test blocks, which is what allows Platt scaling and threshold selection to be performed without any access to test labels. The random-row protocol permits temporally adjacent rows from the same driver and route to fall on both sides of the split and therefore represents the most optimistic case; the time-blocked protocol removes future-to-past information flow within each driver-route sequence; the driver-held-out protocol removes driver identity; and the route-held-out protocol additionally removes the route context, which is the only protocol that tests domain transfer.

Table 4. Validation protocols.

Protocol	Partition rule	Purpose
Random row	64% fit, 16% calibration, 20% test; stratified rows	Shows the conventional estimate and possible adjacency optimism
Time blocked	Within each driver-route sequence: first 64% fit, next 16% calibrate, final 20% test	Prevents future rows from entering model fitting
Driver held out	Five-fold GroupKFold; 3 drivers test, 2 of remaining drivers calibrate, 10 fit	Measures unseen-driver generalization across all routes
Route held out	One whole route test; other two routes split 80% fit/20% temporal calibration	Measures highway, rough-terrain, and urban domain transfer
SC participant held out	Five folds; 4 participants test, 3 calibrate, 13 fit	Prevents overlapping windows from the same participant crossing into test

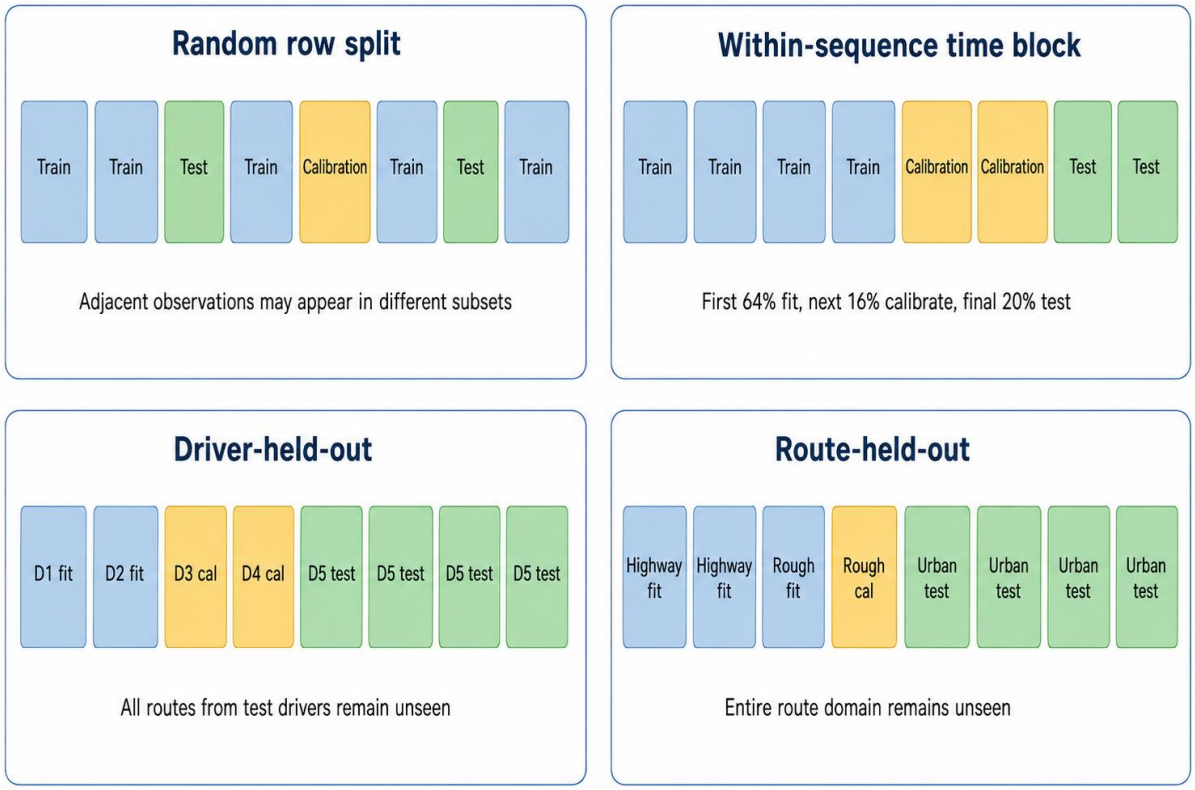


Figure 2. Conceptual comparison of the four DMPS validation protocols. Calibration data are distinct from classifier fitting and final testing.

3.6 Evaluation metrics and statistical analysis

The measures of discrimination were the area under the receiver operating characteristic curve (AUROC) and the area under the precision–recall curve (AUPRC). Due to the limited number of drowsy samples in DMPS, AUPRC was assumed as the primary rank metric. Balanced accuracy, F1, MCC, sensitivity, and specificity were included in the threshold-dependent performance measurements. The quality of the probability map was evaluated by computing the Brier score, negative log likelihood (NLL) and expected calibration error (ECE) within 10 equally sized bins. Better means lower values for Brier, NLL and ECE.

Formally, let TP, FP, TN and FN denote true positives, false positives, true negatives and false negatives at the operating threshold, N the number of test samples, $y_i \in \{0,1\}$ the observed label and $\hat{p}_i$ the calibrated positive-class probability. Precision and recall are defined as

$$P = \frac{\mathrm{TP}}{\mathrm{TP+FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP+FN}} \tag{12}$$

and their harmonic mean gives the F1 score,

$$F_1 = \frac{2PR}{P+R} \tag{13}$$

The primary rank metric is the area under the precision–recall curve,

$$\mathrm{AUPRC} = \int_0^1 P(R)\,\mathrm{d}R \tag{14}$$

which is preferred over AUROC here because the positive prevalence in the pooled DMPS data is only 3.41%. The threshold-dependent agreement measure is

$$\mathrm{MCC} = \frac{\mathrm{TP}\cdot\mathrm{TN}-\mathrm{FP}\cdot\mathrm{FN}}{\sqrt{(\mathrm{TP+FP})(\mathrm{TP+FN})(\mathrm{TN+FP})(\mathrm{TN+FN})}} \tag{15}$$

Probability quality is quantified by the Brier score, the negative log likelihood, and the expected calibration error computed over B = 10 equally sized bins B_b:

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N}(\hat{p}_i - y_i)^2 \tag{16}$$

$$\mathrm{NLL} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \ln \hat{p}_i + (1-y_i)\ln(1-\hat{p}_i)] \tag{17}$$

$$\mathrm{ECE} = \sum_{b=1}^{B}\frac{|B_b|}{N}\left|\mathrm{acc}(B_b) - \mathrm{conf}(B_b)\right| \tag{18}$$

where $\mathrm{acc}(B_b)$ is the empirical positive rate and $\mathrm{conf}(B_b)$ the mean predicted probability within bin B_b. Lower BS, NLL and ECE values indicate better-calibrated probabilities.

The results of the driver- and route-held-out analyses are presented as mean ± sample standard deviation of folds. Driver-clustered nonparametric bootstrap intervals (200 replicates; 15 clusters) were calculated for primary HGB metrics. When performing paired driver-cluster bootstrap differences, identical out-of-fold observations were used as the basis for comparisons between HGB and logistic regression and CQMF. These time-frames reflect uncertainty for the drivers available but shouldn't be read as guarantees for the population as a whole.

3.7 Modality ablation and missing-sensor stress test

There were five combinations of the three modality groups with HGB refitted on each of the non-zero combinations. For the purposes of evaluation of incomplete sensing, the cells of 10%, 20%, 30% or 50% of test feature cells were independently completely masked at random. Masked data was not used to retrain models. HGB applied its country's missing-value routing while CQMF also normalized its quality-adjusted modality weights. This experiment is a stress test, not a true hardware failure model, as actual failures can cause temporal correlation or even the loss of a whole sensor stream.

4. Results

4.1 Dataset audit and domain shift

There was no difference in number of samples collected across the three DMPS routes, but there was a material difference in the prevalence of events. The rate of drowsiness was reduced from 5.06% on the highway to 3.50% on rough terrain and 1.68% in dense urban driving. The opposite was true for sudden braking: 2.04%, 5.55%, and 7.83%, respectively. Therefore, a model can partially infer the context of the route and leverage route specific priors, as long as the route is not held out. The direct brake variables and SC alternate target encodings all had AUROC of 1.000 for their respective outcomes, the latter of which was the only one to have achieved a score of 1.000.

4.2 Main model comparison

Table 5. Main calibrated DMPS results. Values for grouped protocols are mean ± SD across folds.

Protocol	Model	AUROC	AUPRC	F1	MCC	Brier
Random row split	Logistic regression	0.9983	0.9714	0.9120	0.9096	0.00438
Random row split	HGB	0.9991	0.9813	0.9306	0.9287	0.00347
Random row split	CQMF	0.9882	0.8887	0.8169	0.8195	0.00941
Time-blocked split	Logistic regression	0.9942	0.8831	0.4693	0.5359	0.03074
Time-blocked split	HGB	0.9964	0.9204	0.6739	0.6960	0.00937
Time-blocked split	CQMF	0.9911	0.8751	0.8028	0.8008	0.01040
Driver-held-out	Logistic regression	0.9983 ± 0.0004	0.9736 ± 0.0044	0.9165 ± 0.0040	0.9138 ± 0.0043	0.00430 ± 0.00018
Driver-held-out	HGB	0.9990 ± 0.0003	0.9819 ± 0.0041	0.9342 ± 0.0094	0.9319 ± 0.0098	0.00355 ± 0.00039
Driver-held-out	CQMF	0.9918 ± 0.0034	0.9062 ± 0.0271	0.8088 ± 0.0554	0.8054 ± 0.0540	0.01090 ± 0.00572
Route-held-out	Logistic regression	0.9969 ± 0.0003	0.9487 ± 0.0184	0.8181 ± 0.0214	0.8249 ± 0.0178	0.00881 ± 0.00385

Route-held-out	HGB	0.9985 ± 0.0001	0.9686 ± 0.0145	0.8760 ± 0.0401	0.8780 ± 0.0355	0.00527 ± 0.00148
Route-held-out	CQMF	0.9883 ± 0.0006	0.8813 ± 0.0347	0.8059 ± 0.0352	0.8116 ± 0.0280	0.00988 ± 0.00405

Table 5 reports the calibrated results for all three models under the four DMPS protocols. Pooled HGB was the best overall model. Under driver-held-out validation, it achieved AUROC 0.9990 ± 0.0003, AUPRC 0.9819 ± 0.0041, F1 0.9342 ± 0.0094, and MCC 0.9319 ± 0.0098. Driver-clustered 95% bootstrap intervals were 0.99880–0.99919 for AUROC, 0.97938–0.98479 for AUPRC, and 0.00318–0.00393 for Brier score. Relative to logistic regression, HGB increased driver-held-out AUPRC by 0.00933 (95% clustered bootstrap interval 0.00647–0.01210) and AUROC by 0.000744 (0.000503–0.000993).

There was no improvement in the result by CQMF. Its driver-held-out AUPRC was 0.9062 ± 0.0271 and F1 0.8088 ± 0.0554. HGB exceeded CQMF by 0.09160 AUPRC (95% clustered bootstrap interval 0.05703–0.13021). The optimized CQMF weights assigned minimal weight to the vehicle context and were also significantly different for some physiological versus wearable modalities for each driver fold. This finding suggests that the independent modality probabilities were averaged at the end and thus lost some interactions that the tree ensemble that was pooled together captured efficiently.

Only 730 (1.62%) of the time-blocked test observations were positive, while 1,537 (3.42%) of the random test observations were positive. However, HGB's F1 was 0.6739, with AUROC at 0.9964 and AUPRC at 0.9204, but its operating threshold (calibrated from the past high-prevalence block) carried over poorly to the lower-prevalence future block. CQMF achieved slightly lower AUPRC (0.8751) and higher F1 (0.8028) on its independently chosen threshold. The comparison illustrates the need to report rank discrimination, calibration, prevalence, and operating metrics together. Figure 3 plots the route-stratified receiver-operating-characteristic and precision–recall curves obtained from the driver-held-out out-of-fold predictions. The ROC curves are almost indistinguishable across the three routes, whereas the precision–recall curves separate clearly, which is the empirical reason AUPRC rather than AUROC is treated as the primary rank metric in this study.

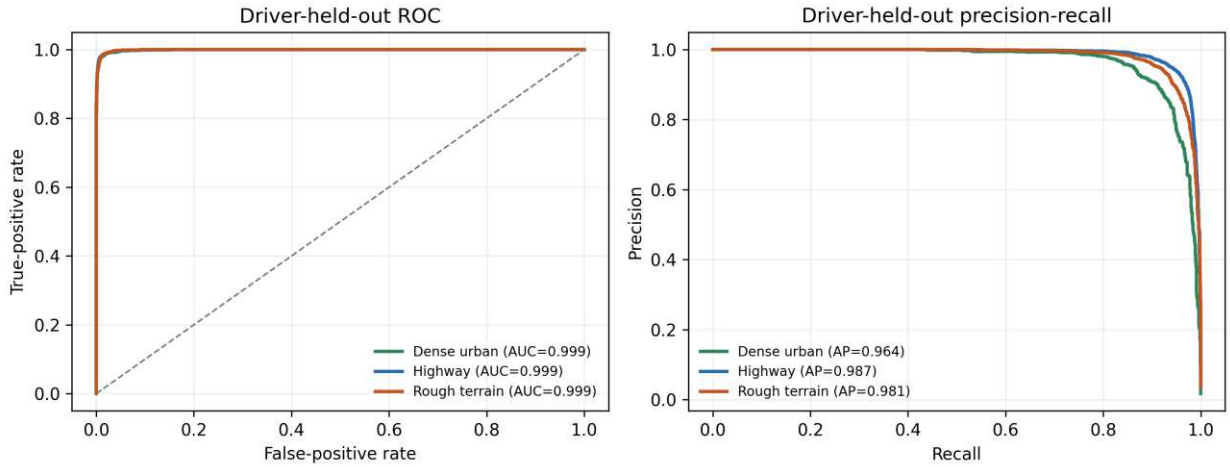


Figure 3. Route-stratified receiver-operating-characteristic and precision–recall curves for pooled HGB using driver-held-out out-of-fold predictions.

4.3 Route-held-out generalization

Table 6. Pooled HGB performance when each route is completely held out.

Held-out route	Positive n	AUROC	AUPRC	Bal. acc.	F1	Sensitivity	Specificity	Brier
Highway	3795	0.9985	0.9798	0.9169	0.9033	0.8345	0.9993	0.00658
Rough terrain	2627	0.9986	0.9736	0.9116	0.8948	0.8238	0.9994	0.00556
Dense urban	1261	0.9985	0.9523	0.8607	0.8299	0.7216	0.9997	0.00366

Table 6 reports pooled HGB performance when each route is withheld in turn. Across the three held-out routes, pooled HGB produced mean AUPRC 0.9686 ± 0.0145 and F1 0.8760 ± 0.0401. The dense-urban route was the most difficult: AUPRC 0.9523, F1 0.8299, and sensitivity 0.7216. Highway performance was stronger (AUPRC 0.9798; sensitivity 0.8345). Specificity exceeded 0.999 on every route, but this should not obscure the lower sensitivity. The route-transfer penalty is operationally important because a fixed alarm threshold could miss more drowsiness events in a new low-prevalence domain.

4.4 Calibration

Driver-held-out pooled HGB yielded Brier score 0.00355 ± 0.00039, NLL 0.01297 ± 0.00142, and ECE 0.00149 ± 0.00121. Under route-held-out validation, these values deteriorated to 0.00527 ± 0.00148, 0.02100 ± 0.00530, and 0.00657 ± 0.00216, respectively. The equal-frequency reliability diagram in Figure 4 displays good match between predicted and observed probabilities following Platt scaling, and the route results reveal that after domain shift calibration needs to be monitored.

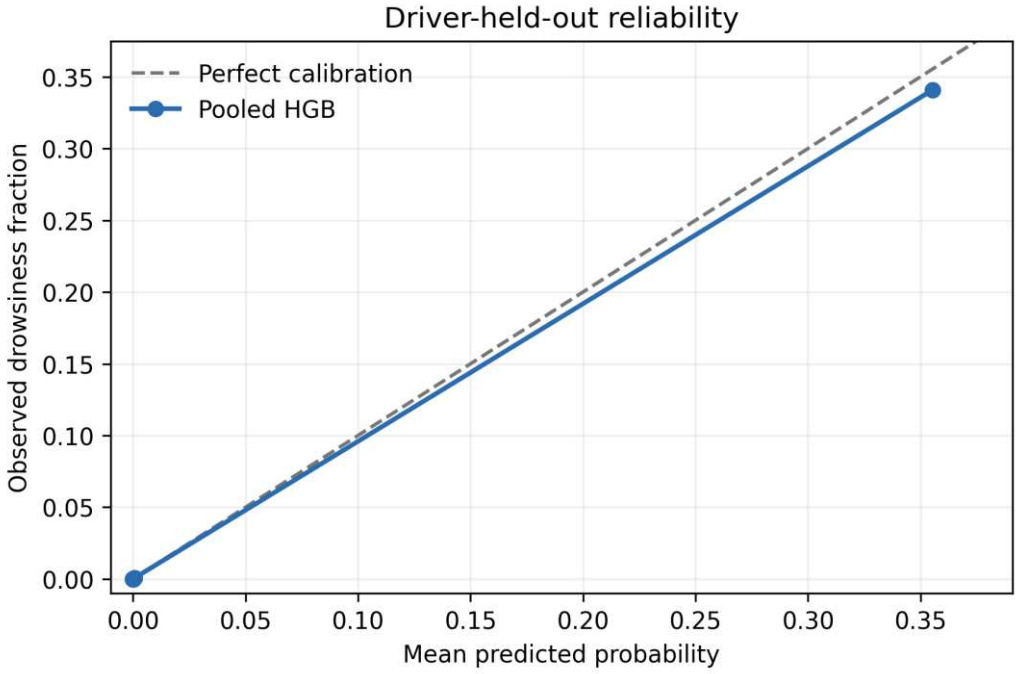


Figure 4. Equal-frequency reliability diagram for pooled HGB under driver-held-out validation. Multiple low-risk bins overlap visually near the origin because drowsiness prevalence is low.

4.5 Modality ablation

Table 7. Pooled HGB modality ablation under driver-held-out validation.

Predictor subset	AUROC	AUPRC	Brier
Physiological	0.9303 ± 0.0174	0.6104 ± 0.0559	0.02635 ± 0.01105
Wearable	0.9863 ± 0.0027	0.8714 ± 0.0187	0.01032 ± 0.00109
Vehicle	0.8973 ± 0.0030	0.2385 ± 0.0077	0.02874 ± 0.00094
Physiological+Wearable	0.9975 ± 0.0006	0.9645 ± 0.0071	0.00522 ± 0.00058
Physiological+Vehicle	0.9786 ± 0.0057	0.7838 ± 0.0426	0.01780 ± 0.00607
Wearable+Vehicle	0.9952 ± 0.0009	0.9307 ± 0.0111	0.00735 ± 0.00090
All pooled	0.9990 ± 0.0003	0.9819 ± 0.0041	0.00355 ± 0.00039

Table 7 reports the ablation over the five non-empty combinations of the three modality groups. Wearable behaviour was the strongest single modality (AUPRC 0.8714), followed by physiology (0.6104) and vehicle context (0.2385). Physiological plus wearable features reached AUPRC 0.9645. Adding vehicle context increased AUPRC to 0.9819, despite the weak vehicle-only result. This pattern supports complementary cross-modal interactions rather than treating each modality as an independently sufficient expert. Figure 5 plots the mean AUPRC and the between-fold standard deviation for each subset, and shows that the gain obtained by adding vehicle context to the physiological–wearable pair is larger than the weak vehicle-only result would predict.

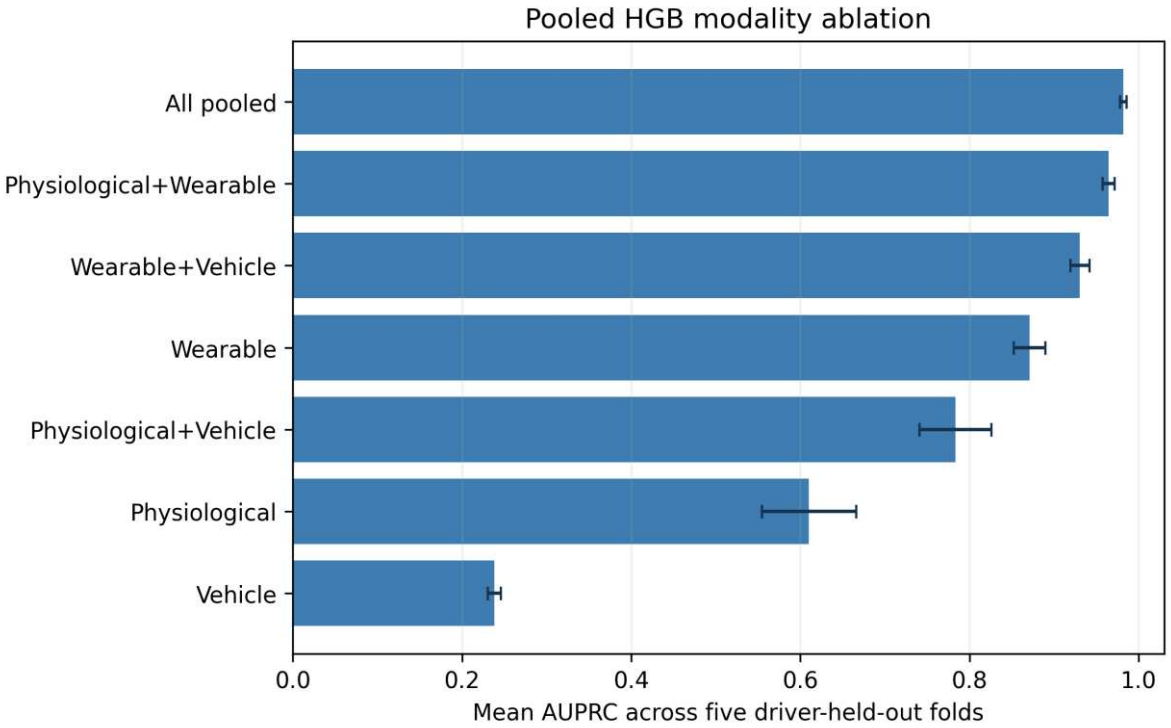


Figure 5. Mean AUPRC and between-fold standard deviation for pooled HGB modality ablations under driver-held-out validation.

4.6 Missing-sensor stress test

Table 8. Missing-sensor stress test under driver-held-out validation.

Missing cells	Model	AUROC	AUPRC	Brier
0%	HGB	0.9990 ± 0.0003	0.9819 ± 0.0041	0.00355 ± 0.00039
0%	CQMF	0.9918 ± 0.0034	0.9062 ± 0.0271	0.01090 ± 0.00572
10%	HGB	0.9953 ± 0.0010	0.9441 ± 0.0097	0.00626 ± 0.00063
10%	CQMF	0.9828 ± 0.0041	0.8440 ± 0.0292	0.01297 ± 0.00450
20%	HGB	0.9885 ± 0.0031	0.8919 ± 0.0223	0.00969 ± 0.00132
20%	CQMF	0.9708 ± 0.0079	0.7766 ± 0.0372	0.01547 ± 0.00402
30%	HGB	0.9784 ± 0.0047	0.8273 ± 0.0228	0.01346 ± 0.00110
30%	CQMF	0.9505 ± 0.0126	0.6953 ± 0.0467	0.01815 ± 0.00324
50%	HGB	0.9409 ± 0.0080	0.6630 ± 0.0219	0.02107 ± 0.00085
50%	CQMF	0.9038 ± 0.0121	0.5260 ± 0.0409	0.02393 ± 0.00214

Table 8 reports both models under progressive random masking of test feature cells. Pooled HGB remained stronger at every masking rate. Its AUPRC declined from 0.9819 without masking to 0.9441 at 10%, 0.8919 at 20%, 0.8273 at 30%, and 0.6630 at 50%. CQMF decreased from 0.9062 to 0.5260 at 50%. Quality weighting therefore did not compensate for the information lost by separate modality learners. The monotonic Brier deterioration for both models indicates that deployment should include sensor-health monitoring and an abstention or fallback policy rather than relying on implicit missing-value handling alone. Figure 6 plots AUPRC against the masking rate with error bars across the five driver-held-out folds, and shows that the margin between pooled HGB and CQMF widens as more cells are removed.



Figure 6. AUPRC under random test-time feature masking. Error bars show standard deviation across five driver-held-out folds.

4.7 Skin-conductance sensitivity analysis

Table 9. HGB sensitivity analysis on the SC workbook after excluding all target proxies.

Protocol	AUROC	AUPRC	F1	Sensitivity	Specificity	Brier
Random row split	0.9982	0.9993	0.9848	0.9854	0.9590	0.01438
Time-blocked split	0.9984	0.9994	0.9863	0.9914	0.9510	0.01371
Participant-held-out	0.9983 ± 0.0001	0.9994 ± 0.0000	0.9848 ± 0.0010	0.9959 ± 0.0039	0.9306 ± 0.0160	0.01449 ± 0.00054

Table 9 reports the SC workbook sensitivity analysis after all target proxies were excluded. After removing KSS and the alternate drowsiness encodings, the random, time-blocked, and participant-held-out SC results remained similar. Participant-held-out AUROC was 0.9983 ± 0.0001, AUPRC 0.99935 ± 0.00004, and F1 0.9848 ± 0.0010. Specificity declined from 0.9590 under random splitting to 0.9306 ± 0.0160 with participants held out, while sensitivity increased to 0.9959 ± 0.0039. These results show strong internal separability of the supplied feature table, but they do not establish naturalistic temporal validity because raw 128-Hz signals, acquisition logs, and an independently reconstructed label timeline were not available.

4.8 Computational cost

Table 10. Computational cost on one driver-held-out fold (150,000 fitting rows; 45,000 test rows).

Model	Fit time (s)	45k inference (s)	µs/sample	Serialized size (kB)
Logistic regression	0.370	0.0120	0.267	2.3
HGB	0.858	0.0434	0.964	274.8
CQMF base learners	1.750	0.1203	2.673	678.8

Table 10 reports fit time, inference time and serialized model size for one driver-held-out fold. On the available CPU runtime, pooled HGB fitted in 0.858 s and predicted 45,000 samples in a median 0.0434 s (0.964 µs per sample). Logistic regression was faster but less accurate, while CQMF required three base learners, approximately doubled fitting time, and used 2.67 µs per sample. These measurements exclude spreadsheet loading, feature acquisition, and communication latency; they characterize only in-memory model execution.

5. Discussion

5.1 Principal findings and validation hierarchy

The main conclusion is methodological and model-dependent. After direct event variables and target proxies were removed, a calibrated pooled HGB model generalized well to unseen drivers and showed high discrimination on unseen routes. The values of route held-out sensitivity and time blocked F1, however, were

materially less than the random or driver held-out summary value. These differences would be masked by one accuracy or AUROC. The results justify a validation hierarchy: (1) random rows as development diagnostic, (2) personalisation claims based on participant held out predictions as minimum evidence, and (3) predictions held out of the route as necessary evidence for operational transfer.

5.2 Why explicit modality fusion did not help

This result also challenges the idea that the more explicit the fusion, the better. CQMF was created to account for the reliability of the modality validation and availability of the sensors, but it performed worse than pooled HGB in full and incomplete sensing. The ablation study explains why. The vehicle variables performed poorly in isolation, yet adding them to the physiological–wearable pair improved performance further when all variables were learned together. Pooled trees can learn conditional rules, for example a physiological pattern that carries a different meaning in a particular vehicle or cabin state. Such interactions cannot be recovered once each modality has been averaged at the probability level. The negative result is practically useful because in recent multimodal work, more and more attention is placed on architectural coupling [9,10] without considering whether the leakage-controlled baseline, based on a simpler pooled learner, is strong enough.

5.3 Relation to the authors' prior work

This analysis brings together the ideas of the four source papers without repeating their findings. It retains physiology, grip, limb and vehicle variables from the wearable monitoring system [1]. From VigilEye [2] it retains the premise that non-contact behavioural information is complementary, although no facial images were present in the provided workbooks. It inherits SC decomposition, context, and adaptive probability interpretation from Deep-ATM [3] and adopts participant-held-out testing, replacing the segmentation by participants with the segmentation by the algorithm. It retains driver–vehicle–environment context and introduces route-held-out generalization, as inherited from DrivNet [4]. The key novelty is actually the evaluation framework and the confirmed results as opposed to the re-labeling of the previous architectures.

5.4 Calibration, thresholds and deployment implications

The calibration was good for drivers that were not observed and bad for a route shift. This is important because the best alert threshold will be influenced by the prevalence of the condition, the tolerance for false alarms and the cost of missing an event of drowsy driving. It is evident from the results of the time-blocked test that even a good model with high AUPRC may suffer a drop in F1 when the future positive rate is not similar to the calibration rate. A deployable system should thus record the calibration drift, re-evaluate the thresholds according to the routes and operating contexts, and establish an abstention and/or escalation state. It shouldn't give an alarm if a good ranking score is converted silently on a threshold optimised for a previous prevalence.

Pooled HGB is computationally light, allowing for the use of an edge deployment, but the inference benchmark excludes sensor sampling, filtering, synchronization, wireless transfer, and alarm logic. The missingness experiment also demonstrates that missingness support in native is not the same as fault tolerance. HGB AUPRC decreased by about 0.155 when there were missing cells of 30%. The practical implementation should display the sensor-health indicators, detect missing channels and gracefully degrade. In other words, the system might continue functioning normally when the amount of missing data is low, seek recalibration when it is in the middle range and abstain or change to a camera-based backup at high missingness.

5.5 Limitations

Author supplied feature workbooks were used to analyse the data, not the raw synchronized sensor streams. No acquisition logs or hardware timestamps were provided and there were no traces of rejection of artifacts which would allow for an independent reconstruction of the label timeline.

The DMPS data contain only 15 distinct drivers and the SC data only 20 distinct participants. Therefore, when applied to large row counts, cluster uncertainty is based on small numbers of people.

The very high HGB performance indicates that the supplied feature tables are strongly separable. These results should not be read as a prediction of field performance before independent replication from raw data.

Only three types of routes were validated with route held out. Differences in geographic, cultural, vehicle, weather, shift-work, medication and sleep-history were not represented.

Same-time detection was evaluated in the study. No validated pre-event horizon could be established and claimed from the data.

Random cell masking assumes that the data is missing completely at random. True failures can involve the complete sensor, last for extended periods or be associated with vibration, weather or driver movement.

Facial video, EEG and EOG were not present. The framework can thus not compare all the clinically and operationally relevant modalities.

The current study is not conducting prospective alarm-utility testing, human-factors assessment, false alarms/per driving hour, or intervention effectiveness.

5.6 Future work

The next study should focus on collecting raw physiological and grip data (and possibly vehicle and video) and be synchronized with immutable timestamps, pre-registering driver-, session-, route-level splits before window extraction, and establishing a clinically or operationally meaningful prediction horizon. External validation on public continuous multimodal resource like UL-DD [11] would validate transportability. False alarms per hour, missed event cost, calibration drift, and driver acceptance are factors to consider when evaluating threshold policies. Lastly, structured sensor outages should be tested in training and abstention should be treated as a safety operation, not a requirement to make a prediction from very limited data.

6. Conclusion

In this study, a leakage-resistant evaluation framework for multimodal driver drowsiness detection is presented, involving 225,000 driver–route observations and 87,430 skin-conductance windows. Target-derived proxies, identity-related features, and other potential sources of information leakage are eliminated, resulting in a more accurate evaluation of model performance when randomly sampling rows, time-blocking, driver-holding out, and route-holding out.

The approach with best and consistent performance during the evaluation was the calibrated pooled histogram-based gradient boosting approach. Under five-fold driver-held-out validation, the model achieved an AUROC of 0.9990 ± 0.0003, AUPRC of 0.9819 ± 0.0041, F1 score of 0.9342 ± 0.0094, MCC of 0.9319 ± 0.0098, and Brier score of 0.00355 ± 0.00039. The analysis of the modalities further indicated that a combination of physiological, wearable-behaviour and vehicle-context information provided the best performance. To the contrary, there was no benefit from the proposed quality-weighted modality-fusion approach, indicating that late probability fusion may be losing the cross-modal interaction that is well captured by pooled nonlinear learning.

Importantly, strong unseen-driver discrimination did not eliminate deployment uncertainty. The mean AUPRC level dropped to 0.9686 and mean sensitivity dropped to 0.7933 in the case of excluding whole driving routes from the model development, where the dense-urban route most significantly reduced the sensitivity. Moreover, as the number of missing sensors increased, the performance of the HGB decreased, with the pooled HGB AUPRC dropping from 0.9819 when there were no missing sensors to 0.6630 when 50% of the features were lost at random. The results confirm that, with standard discrimination measure staying high, route-domain shift, changes in prevalence, calibration drift, and sensor availability can significantly influence operational performance.

Removing alternate drowsiness encodings also left the SC sensitivity analysis with a high held-out separability. But with no synchronized raw acquisition streams, no label timelines that could be reconstructed independently, and no prospective external validation, these results cannot be considered as proof of actual early-warning performance in the real world. Therefore, the present results are supportive of calibrated pooled multimodal modeling for same-time drowsiness detection and not of prospective prediction or deployment readiness.

The main contribution of this work lies not just on the identification of a high-performing classifier, but the proof that credible multimodal drowsiness assessment necessitates the explicit leakage auditing, participant-disjoint evaluation, route-level generalization testing, independent probability calibration, sensor-failure assessment, and transparent data provenance. Before the framework can be applied to real-world driver monitoring systems, future prospective studies using synchronized raw multimodal signals, pre-specified driver/session/route splits, independent external datasets, etc., and safety-oriented measures like false alarms per driving hour, missed-event cost etc. are necessary.

Declarations

Ethics approval and consent to participate

This study was a retrospective secondary analysis of de-identified datasets derived from the authors' previously published studies and involved no new participant recruitment, intervention or collection of identifiable information. The original DMPS study involved 15 willing participants who signed a written agreement confirming their willingness and ability to participate before commencement of the experiment. The original skin-conductance study was approved by the Ethics Review Committee of Vel Tech Rangarajan Dr Sagunthala R&D Institute of Science and Technology under approval number VT/ERC/2022-104. All procedures for the skin-conductance study were performed in accordance with relevant institutional guidelines and regulations and the Declaration of Helsinki. Written informed consent was obtained from all participants before the original data collection.

Consent for publication: Written informed consent for participation and for publication of anonymized data was obtained from participants in the original skin-conductance study. The present secondary analysis contains no participant names or directly identifying tabular information.

Funding: The authors received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors for this secondary analysis.

Competing interests: The authors declare no competing interests.

Data availability: The DMPS and skin-conductance workbooks analysed in this study were derived from the authors' previously published experimental studies. The datasets are not deposited in a public repository. De-identified processed workbooks may be obtained from the corresponding author upon reasonable request,

subject to the ethical, consent and institutional data-governance conditions applicable to the original studies. Raw synchronized sensor streams were not available for the present secondary analysis.

Code availability: The analysis code, fixed model configurations, data-partitioning procedures, fold-level predictions and figure-generation scripts are available from the corresponding author on reasonable request. The code does not contain participant identifiers or raw sensor records.

Author contributions: J.R.T. contributed to conceptualization, methodology, software development, formal analysis, investigation, data curation, visualization, and preparation of the original draft. S.P. contributed to conceptualization, methodology, validation, supervision, resources, and review and editing of the manuscript. R.D. contributed to validation, investigation, data interpretation, and review and editing. R.P. contributed to methodology, validation, supervision, project administration, and review and editing. All authors reviewed and approved the final manuscript.

References

1. Ponnan S, Theivadas JR, HemaKumar VS, Einarson D. Driver monitoring and passenger interaction system using wearable device in intelligent vehicle. Computers & Electrical Engineering. 2022;103:108323. https://doi.org/10.1016/j.compeleceng.2022.108323.

2. Theivadas JR, Ponnan S. VigilEye: Machine learning-powered driver fatigue recognition for safer roads. Measurement: Sensors. 2024;33:101186. https://doi.org/10.1016/j.measen.2024.101186.

3. Theivadas JR, Ponnan S. Deep-ATM DL-LSTM: A novel adaptive thresholding model with dual-layer LSTM architecture for real-time driver drowsiness detection using skin conductance signals. Computers in Biology and Medicine. 2025;192:110243. https://doi.org/10.1016/j.compbiomed.2025.110243.

4. Theivadas JR, Ponnan S. DrivNet: Adaptive driving behavior prediction for improving the safety of multimodal transportation in autonomous vehicle systems. Multimodal Transportation. 2026;5:100266. https://doi.org/10.1016/j.multra.2025.100266.

5. Hasan MM, Watling CN, Larue GS. Physiological signal-based drowsiness detection using machine learning: Singular and hybrid signal approaches. Journal of Safety Research. 2022;80:215–225. https://doi.org/10.1016/j.jsr.2021.12.001.

6. Amidei A, Spinsante S, Iadarola G, Benatti S, Tramarin F, Pavan P, Rovati L. Driver drowsiness detection: A machine learning approach on skin conductance. Sensors. 2023;23(8):4004. https://doi.org/10.3390/s23084004.

7. Hasan MM, Watling CN, Larue GS. Validation and interpretation of a multimodal drowsiness detection system using explainable machine learning. Computer Methods and Programs in Biomedicine. 2024;243:107925. https://doi.org/10.1016/j.cmpb.2023.107925.

8. Schwarz C, Gaspar J, Yousefian R. Multi-sensor driver monitoring for drowsiness prediction. Traffic Injury Prevention. 2023;24(sup1):S100–S104. https://doi.org/10.1080/15389588.2023.2164839.

9. Hu S, Gao Q, Xie K, et al. Efficient detection of driver fatigue state based on all-weather illumination scenarios. Scientific Reports. 2024;14:17075. https://doi.org/10.1038/s41598-024-67131-5.

10. Cao S, Feng P, Kang W, Chen Z, Wang B, et al. Optimized driver fatigue detection method using multimodal neural networks. Scientific Reports. 2025;15:12240. https://doi.org/10.1038/s41598-025-86709-1.

11. Bodaghi M, Hosseini M, Gottumukkala R, Bhupatiraju RT, Ahmad I, Gabbouj M. A multimodal drowsiness dataset using video, biometric, and behavioral data. Scientific Data. 2026;13:506. https://doi.org/10.1038/s41597-025-06540-1.

12. Nakagama Y, Ishii D, Yoshizoe K. Comparison of lightweight methods for vehicle dynamics-based driver drowsiness detection. arXiv. 2025;arXiv:2506.07014. https://doi.org/10.48550/arXiv.2506.07014.

13. AlArnaout Z, Zaki C, Kotb Y, AlAkkoumi M, et al. Exploiting heart rate variability for driver drowsiness detection using wearable sensors and machine learning. Scientific Reports. 2025;15:24898. https://doi.org/10.1038/s41598-025-08582-2.

14. Kaida K, Takahashi M, Åkerstedt T, Nakata A, Otsuka Y, Haratani T, Fukasawa K. Validation of the Karolinska Sleepiness Scale against performance and EEG variables. Clinical Neurophysiology. 2006;117(7):1574–1581. https://doi.org/10.1016/j.clinph.2006.03.011.

15. Ghoddoosian R, Galib M, Athitsos V. A realistic dataset and baseline temporal model for early drowsiness detection. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops. 2019. https://doi.org/10.1109/CVPRW.2019.00027.

16. Bajaj JS, Kumar N, Kaushal RK, Gururaj HL, Flammini F, Natarajan R. System and method for driver drowsiness detection using behavioral and sensor-based physiological measures. Sensors. 2023;23(3):1292. https://doi.org/10.3390/s23031292.

17. Friedman JH. Greedy function approximation: A gradient boosting machine. Annals of Statistics. 2001;29(5):1189–1232. https://doi.org/10.1214/aos/1013203451.

18. Guo C, Pleiss G, Sun Y, Weinberger KQ. On calibration of modern neural networks. Proceedings of Machine Learning Research. 2017;70:1321–1330.

19. Saito T, Rehmsmeier M. The precision–recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLOS ONE. 2015;10(3):e0118432. https://doi.org/10.1371/journal.pone.0118432.

20. Brier GW. Verification of forecasts expressed in terms of probability. Monthly Weather Review. 1950;78(1):1–3. https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.

21. Persson A, Jonasson H, Fredriksson I, Wiklund U, Ahlström C. Heart rate variability for classification of alert versus sleep deprived drivers in real road driving conditions. IEEE Transactions on Intelligent Transportation Systems. 2021;22(6):3316–3325. https://doi.org/10.1109/TITS.2020.2981941.